Learning Interactions in Collective Dynamics

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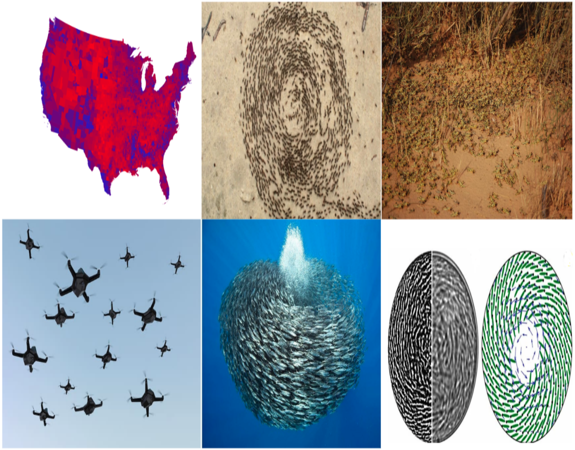
Outline

Collective dynamics

[1] (Enkeleida Lushi, Hugo Wioland, and Raymond E Goldstein. "Fluid flows created by swimming bacteria drive self-organization in confined suspensions". In: *Proceedings of the National Academy of Sciences* 111.27 [2014], pp. 9733–9738)

Collective dynamics



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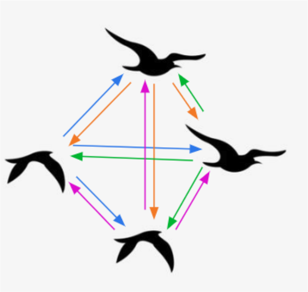
[1] (Lushi, Wioland, and Goldstein, "Fluid flows created by swimming bacteria drive self-organization in confined suspensions")

Collective dynamics definition

Collective dynamics / Collective behavior

It is a process where global order emerges from an initially disordered system via local interactions [2]



[2] (W Ross Ashby. "Principles of the self-organizing dynamic system". In: *The Journal of general psychology* 37.2 [1947], pp. 125–128)

Collective dynamics in various fields and its applications

- Examples of collective behavior in different fields:
  - Physics: Spontaneous magnetization, crystal growth, superconductivity
  - Chemistry: Molecular self-assembly, self-assembled monolayers
  - Biology: Formation of lipid bilayer membranes, social behavior in insects, flocking of birds
  - Human Society: Market economy, traffic dynamics

[3] (Matt Simon. *These Adorable Fish Robots Form Schools Like the Real Thing*. Accessed: 2025-01-13. 2025. URL: https://www.wired.com/story/these-adorable-fish-robots-form-schools-like-the-real-thing/)

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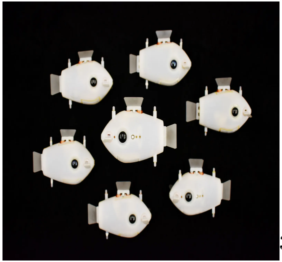
  - Chemistry: Molecular self-assembly, self-assembled monolayers
  - Biology: Formation of lipid bilayer membranes, social behavior in insects, flocking of birds
  - Human Society: Market economy, traffic dynamics
- Applications:
  - Particle swarm optimization
  - Consensus analysis and social behavior
  - Animal migration and conservation
  - Unmanned aerial vehicles (UAV) and self-driving cars

[3]

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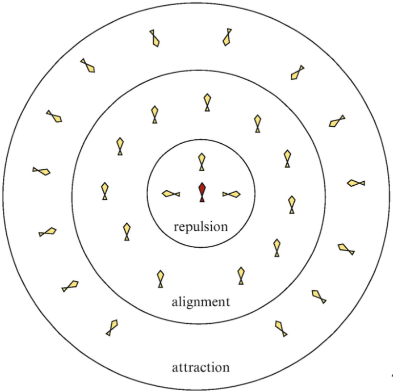
Modeling interactions

- In 1987, Craig Reynolds developed the first influential self-organizing framework for modeling collective motion, characterized by three key principles.



[4]

[4] (Roman Shvydkoy. "Forced Systems". In: *Dynamics and Analysis of Alignment Models of Collective Behavior*. Nečas Center Series. Birkhäuser, Cham, 2021. DOI: 10.1007/978-3-030-68147-0_3. URL: https://doi.org/10.1007/978-3-030-68147-0_3)

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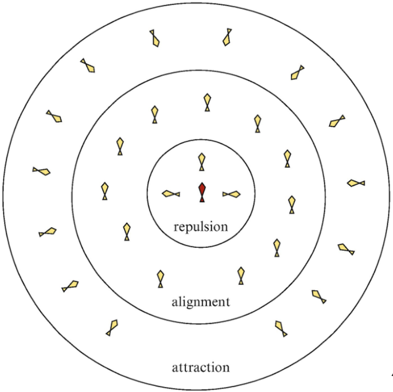

[4]

Subsequent modeling approaches building on Reynolds' flocking model:

- Self-Propelled Particle (SPP) model
- Cucker–Smale model
- Krause model
- Tadmor model
- Kuramoto model

[4] (Shvydkoy, "Forced Systems")

Forward modeling

Consider a system of N agents, where each agent i has a state vector $x_i(t) \in \mathbb{R}^d$.

- **First-Order: Energy-Based Model**

$$\dot{x}_i(t) = f_{\text{env}}^x(x_i) + \frac{1}{N}\sum_{j=1}^{N} \phi_x(|x_j - x_i|)(x_j - x_i)$$

- **Second-Order: Energy–Alignment Model**

$$\dot{x}_i(t) = v_i$$

$$\dot{v}_i(t) = f_{\text{env}}^v(x_i, v_i) + \frac{1}{N}\sum_{j=1}^{N} \Big(\phi_x(|x_j - x_i|)(x_j - x_i) + \phi_v(|x_j - x_i|)(v_j - v_i)\Big)$$

Notation:

- f_{env}: environmental forcing term
- $\phi_x : \mathbb{R}^+ \to \mathbb{R}$ — energy-based interaction law (repulsion/attraction)
- $\phi_v : \mathbb{R}^+ \to \mathbb{R}$ — alignment-based interaction law

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Interaction Kernel

ϕ defines the dynamics of the collective system

General mathematical framework

Commonality

- All models assume **pairwise interactions**.

Why? In collective dynamics, it is reasonable to assume that the influence on any given agent is the sum of the influences exerted by all other agents.

Objective

The goal is to learn the interaction kernel ϕ and the environmental force f_{env} from observed trajectory data, **without assuming a specific analytical form**.

More precisely

- Data $\{x_{data}, \dot{x}_{data}\} = \{x_i^{(m)}(t_l), \dot{x}_i^{(m)}(t_l)\}_{i=1,l=0,m=1}^{N,L,M}$

$$
\left[\left[\begin{array}{c|ccccc}
 & t_0 & \cdots & t_l & \cdots & t_L \\
\hline
1 & x_1(t_0) & \cdots & x_1(t_l) & \cdots & x_1(t_L) \\
\vdots & \vdots & \ddots & \vdots & \cdots & \vdots \\
i & x_i(t_0) & \cdots & x_i(t_l) & \cdots & x_i(t_L) \\
\vdots & \vdots & \ddots & \vdots & \cdots & \vdots \\
N & x_N(t_0) & \cdots & x_N(t_l) & \cdots & x_N(t_L)
\end{array}\right]_{m=1} \right]_{\dots} \Bigg]_{m=M}
$$

More precisely

- Data $\{x_{data}, \dot{x}_{data}\} = \{x_i^{(m)}(t_l), \dot{x}_i^{(m)}(t_l)\}_{i=1,l=0,m=1}^{N,L,M}$

$$\begin{array}{cc} & \begin{array}{ccccc} t_0 & \cdots & t_l & \cdots & t_L \end{array} \\ \begin{array}{c} 1 \\ \vdots \\ i \\ \vdots \\ N \end{array} & \left[\left[\begin{bmatrix} x_1(t_0) & \cdots & x_1(t_l) & \cdots & x_1(t_L) \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ x_i(t_0) & \cdots & x_i(t_l) & \cdots & x_i(t_L) \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ x_N(t_0) & \cdots & x_N(t_l) & \cdots & x_N(t_L) \end{bmatrix}_{m=1}\right]_{\dots}\right]_{m=M} \end{array}$$

- Model

$$\dot{x}_i(t) = f_{\text{env}}^x(x_i) + \frac{1}{N}\sum_{j=1}^{N} \phi_x(|x_j - x_i|)(x_j - x_i)$$

More precisely

- Data $\{x_{data}, \dot{x}_{data}\} = \{x_i^{(m)}(t_l), \dot{x}_i^{(m)}(t_l)\}_{i=1,l=0,m=1}^{N,L,M}$

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\vdots & \ddots & \vdots & \cdots & \vdots \\
x_i(t_0) & \cdots & x_i(t_l) & \cdots & x_i(t_L) \\
\vdots & \ddots & \vdots & \cdots & \vdots \\
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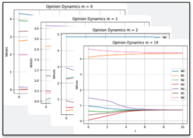
$$

- Model

$$\dot{x}_i(t) = f_{\text{env}}^x(x_i) + \frac{1}{N}\sum_{j=1}^{N} \phi_x(|x_j - x_i|)(x_j - x_i)$$

- Objective



$+$ $\boxed{\dot{x}_i(t) = f_{\text{env}}^x(x_i) + \frac{1}{N}\sum_{j=1}^{N} \phi_x(|x_j - x_i|)(x_j - x_i)}$ Model $\Rightarrow$ ϕ, **f**

Nonparametric Estimator

When the only unknown in the system is the interaction kernel, an algorithm for nonparametric inference of interaction laws was introduced by Mauro Maggioni and colleagues (2019).

PNAS

Nonparametric inference of interaction laws in systems of agents from trajectory data

Fei Lu[a,b,c], Ming Zhong[b], Sui Tang[a], and Mauro Maggioni[a,b,c,d,1]

[a]Department of Mathematics, Johns Hopkins University, Baltimore, MD 21218; [b]Department of Applied Mathematics and Statistics, Johns Hopkins University, Baltimore, MD 21218; [c]Institute for Data Intensive Engineering and Science, Johns Hopkins University, Baltimore, MD 21218; and [d]Mathematical Institute for Data Science, Johns Hopkins University, Baltimore, MD 21218

Edited by Bin Yu, University of California, Berkeley, CA, and approved June 3, 2019 (received for review December 26, 2018)

Data-driven Discovery of Emergent Behaviors in Collective Dynamics

Mauro Maggioni[a, b], Jason Miller[a], and Ming Zhong[a]

[a]Department of Applied Mathematics & Statistics
[b]Department of Mathematics
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Our contributions:

- Extend the algorithm to learn both the f_{env} term and the ϕ term.
- Consider two cases for f_{env}:
  - **Parametric:** informed by scientific knowledge of its form.
  - **Nonparametric:** learned purely from data.

Parametric estimation: inference of f_{env}

Given the data $\{x_{t_l}^{(m)}, \dot{x}_{t_l}^{(m)}\}_{l=0,m=1}^{L,M}$ with $0 = t_0 < t_1 < \cdots < t_L = T$ and $x_0^{(m)} \sim \mu^N(\mathbb{R}^D)$,

$$\dot{x}_i^{(m)} = f_{\text{env}}^x(x_i^{(m)}, \Theta) + \frac{1}{N}\sum_{j=1}^{N} \phi(|x_j^{(m)} - x_i^{(m)}|)(x_j^{(m)} - x_i^{(m)})$$

$$:= f_{\text{env}}^x(x_i^{(m)}, \Theta) + f_\phi(x_i^{(m)}).$$

Empirical error functional: For any φ and θ, define

$$\varepsilon_{L,M}(\varphi, \theta) := \frac{1}{LM}\sum_{l=0}^{L}\sum_{m=1}^{M} \left\| \dot{x}_{t_l}^{(m)} - f_{\text{env}}^x(x_{t_l}^{(m)}, \theta) - f_\varphi(x_{t_l}^{(m)}) \right\|_S^2.$$

Estimator: the minimizer of $\varepsilon_{L,M}$ over $\varphi \in \mathcal{H}$ and $\theta \in \mathcal{K} \subseteq \mathbb{R}^k$ for some compact set $\mathcal{K}$:

$$\hat{\phi}, \hat{\Theta} := \arg \min_{\varphi \in \mathcal{H}, \theta \in \mathcal{K}} \varepsilon_{L,M}(\varphi, \theta).$$

$$|x_{t_l}|_S^2 := \frac{1}{N}\sum_{i=1}^{N} \|x_i(t_l)\|_{l^2(\mathbb{R}^D)}^2.$$

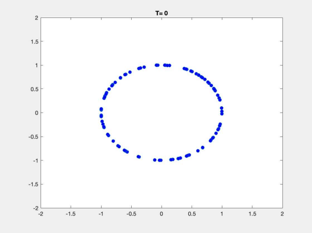
1st-Order system – Kuramoto

This model describs synchronization in a system of coupled oscillators.

$$\dot{\theta}_i = \omega_i + \frac{1}{N} \sum_{j=1}^{N} K_{ij} \sin(\theta_j - \theta_i)$$

where

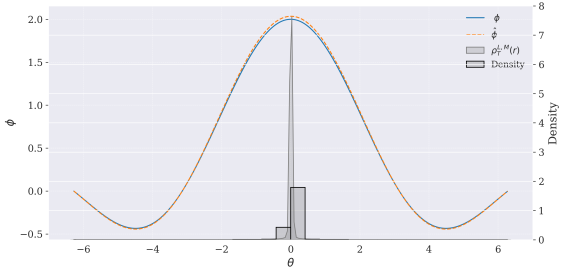
- N – number of oscillators
- θ_i – phase of oscillator i
- K_{ij} – coupling constant
- ω_i – intrinsic frequency
- Coupling strength is constant across all oscillators

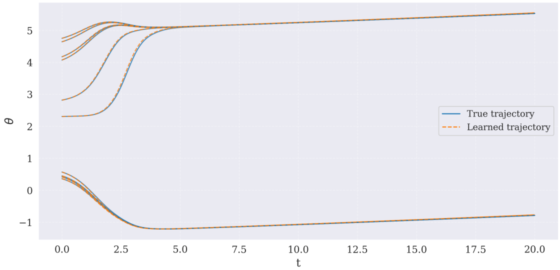


Kuramoto system learning

For all agents i, $\omega_i = \theta_i/100$ and $K_{ij} = K$.

$$\dot{\theta}_i = \underbrace{\omega_i}_{f^x_{\text{env}}} + \frac{1}{N}\sum_{j=1}^{N} \underbrace{K\,\frac{\sin(\theta_j - \theta_i)}{\theta_j - \theta_i}}_{\phi}(\theta_j - \theta_i)$$





Parameter learning

Learning algorithm parameters are as follows:

$$\Theta = [0.01]$$

$$\hat{\Theta} = [0.010156]$$

M	L	t_0	T	N	n	d	ω_i	K
20	200	0	20	10	20	3	$\theta_i/100$	2

$$f^x_{\text{env}} = \theta_i/100$$

2nd-Order system - Cucker–Smale

It describes collective systems like flocking with velocity alignment.

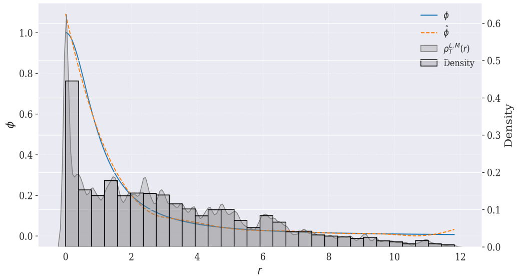
$$\dot{x}_i = v_i,$$
$$\dot{v}_i = f^v_{\text{env}}(x_i, v_i) + \frac{1}{N}\sum_{j=1}^{N} \phi_v(\|x_j - x_i\|)(v_j - v_i).$$

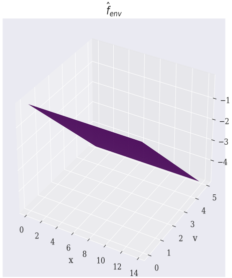
$$f^v_{\text{env}}(x_i, v_i) = -\gamma v_i,$$
$$\phi_v(r) = \frac{1}{(1+r^2)^\beta}.$$

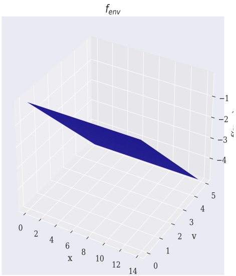
Algorithm Parameters:

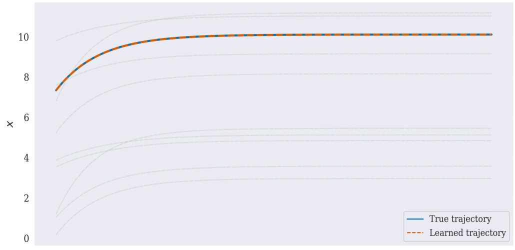
M	L	N	n_r	d_r	$u_0 = [x_0, v_0]$	γ	β
20	1001	10	10	2	$[\,x_0 \sim \mathcal{U}(0.1, 10),\ v_0 \sim \mathcal{U}(0.1, 5)\,]$	1.0	1.0

Cucker–Smale learning









$$\Theta = [-\gamma] = [-1]$$

$$\hat{\Theta} = [-1.000007]$$

2nd-order system – Self-Propelled Particle (SPP) system

This system describes the swarming behavior of multi-agent systems with self-propulsion:

$$\dot{x}_i = v_i$$
$$\dot{v}_i = \left[(\alpha - \beta|v_i|^2)v_i - \nabla_i U(x_i)\right]$$

with the generalized Morse potential given by

$$U(x_i) = \sum_{j \neq i}^{N} \left(C_r e^{-\|x_i - x_j\|/l_r} - C_a e^{-\|x_i - x_j\|/l_a} \right).$$

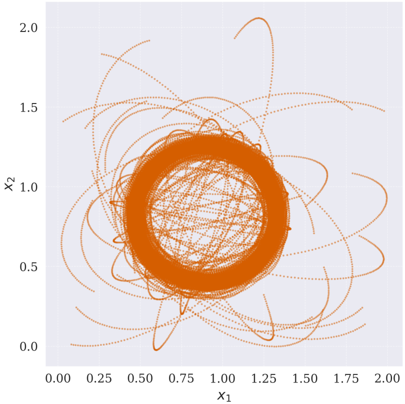
where

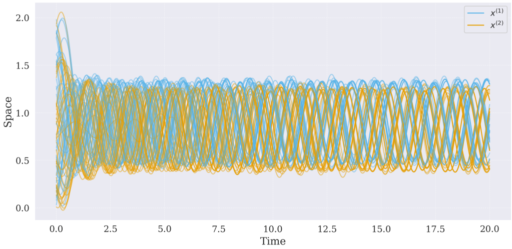
- α – self-propulsion strength
- β – nonlinear damping
- l_a, l_r – attractive and repulsive potential ranges, respectively
- C_a, C_r – amplitudes of the attractive and repulsive potentials, respectively

Simulation – Clump formation

Model parameters used in the simulation:

α	β	C_r	l_r	C_a	l_a
1	0.5	0.6	0.5	1	1





SPP learning

Let $r_{ij} = x_j - x_i$. The system can be written as:

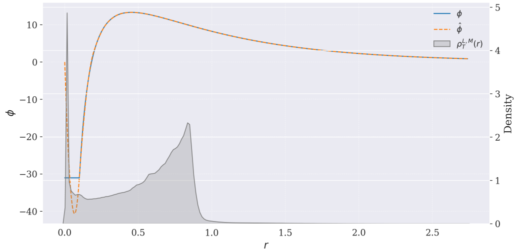
$$\dot{x}_i = v_i$$

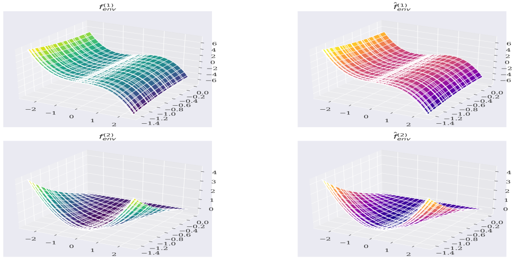
$$\dot{v}_i = \underbrace{(\alpha - \beta \|v_i\|^2) v_i}_{f^x_{\text{env}}} + \sum_{\substack{j=1 \\ j \neq i}}^{N} \left(\underbrace{\frac{C_r}{l_r} e^{-\|r_{ij}\|/l_r} - \frac{C_a}{l_a} e^{-\|r_{ij}\|/l_a}}_{\phi} \right) \frac{r_{ij}}{\|r_{ij}\|}$$

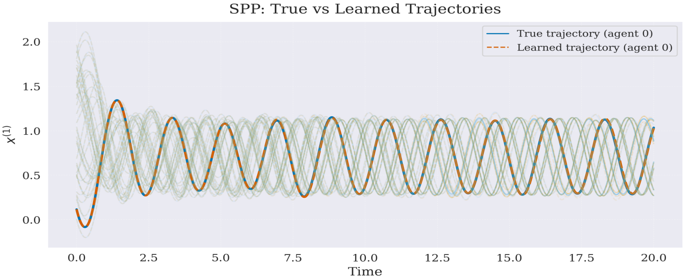
Learning algorithm parameters:

M	L	t_0	T	N	n_r	d_r
20	2000	0	20	40	30	2

SPP learning







$$\Theta = [\alpha, -\beta] = [1, -0.5]$$

$$\hat{\Theta} = [0.999684, -0.499966]$$

$$\dot{x}_i^{(m)} = f_{\text{env}}^x(x_i^{(m)}) + \frac{1}{N}\sum_{j=1}^{N} \phi\big(\|x_j^{(m)} - x_i^{(m)}\|\big)\,(x_j^{(m)} - x_i^{(m)})$$

$$:= f_{\text{env}}(x_i^{(m)}) + f_\phi(x_i^{(m)})$$

Empirical error functional: For any $\varphi_{\text{coll}} \in \mathcal{H}_{\text{coll}}$ and $\varphi_{f_{\text{env}}} \in \mathcal{H}_{f_{\text{env}}}$,

$$\varepsilon_{L,M}\big(\varphi_{\text{coll}}, \varphi_{f_{\text{env}}}\big) := \frac{1}{LM}\sum_{l=0}^{L}\sum_{m=1}^{M} \Big\| \dot{x}_{t_l}^{(m)} - \varphi_{f_{\text{env}}}(x_{t_l}^{(m)}) - f_{\varphi_{\text{coll}}}(x_{t_l}^{(m)}) \Big\|_S^2$$

Estimator: The estimators $\hat{\phi}, \hat{f}_{\text{env}}$ are defined as

$$\big(\hat{\phi}, \hat{f}_{\text{env}}\big)_{L,M,\mathcal{H}} := \underset{\substack{\varphi_{\text{coll}} \in \mathcal{H}_{\text{coll}} \\ \varphi_{f_{\text{env}}} \in \mathcal{H}_{f_{\text{env}}}}}{\arg\min}\ \varepsilon_{L,M}\big(\varphi_{\text{coll}}, \varphi_{f_{\text{env}}}\big)$$

- $\mathcal{H}_{\text{coll}}$: functions $\varphi_{\text{coll}} : \mathbb{R}_+ \to \mathbb{R}$ (interaction kernels)
- $\mathcal{H}_{f_{\text{env}}}$: functions $\varphi_{f_{\text{env}}} : \mathbb{R}^d \to \mathbb{R}^d$ (environmental forces)

$$|x_{t_l}|_S^2 : \frac{1}{N}\sum_{i=1}^{N} |x_i(t_l)|_{l^2(\mathbb{R}^D)}^2$$

Learning Parameters

- **Kuramoto**

M	L	N	n_r	n_x	d_r	d_x	$u_0 = [x_0]$
20	200	10	20	10	3	1	$x_0^{(m,i)} \sim \mathcal{U}(0, 2\pi)$

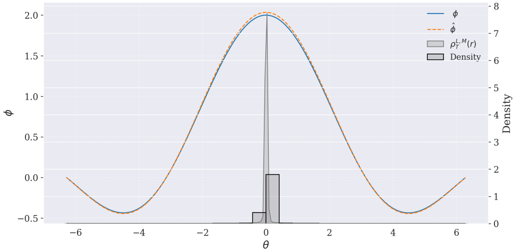
- **Cucker–Smale**

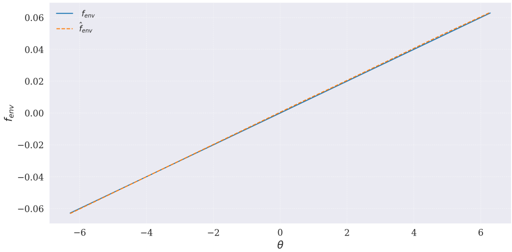
M	L	N	n_r	n_x	n_v	d_r	d_x	d_v	$u_0 = [x_0, v_0]$
20	1001	10	10	10	10	2	1	2	$[\, x_0^{(m,i)} \sim \mathcal{U}(0.1, 10),\ v_0^{(m,i)} \sim r_i \hat{d}_i \,]$

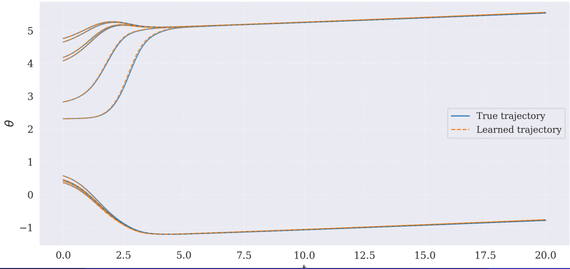
- **Self-Propelled Particle (SPP)**

M	L	N	n_r	n_x	n_v	d_r	d_x	d_v	$u_0 = [x_0, v_0]$
20	1001	10	30	10	30	2	1	2	$[\, x_0^{(m,i)} \sim \mathcal{U}([0, L]^2),\ v_0^{(m,i)} \sim r_i \hat{d}_i \,]$

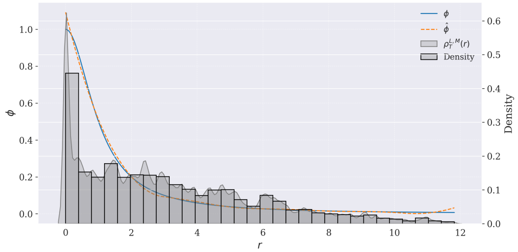
Kuramoto learning: non parametric

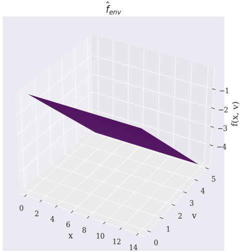


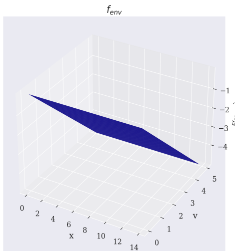


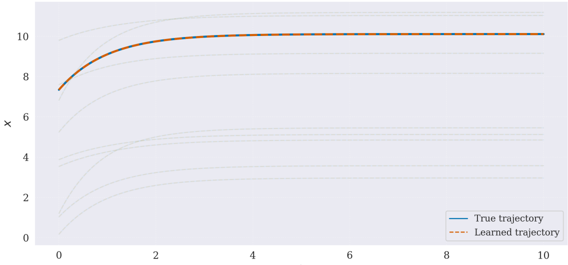


Cucker–Smale learning: non parametric

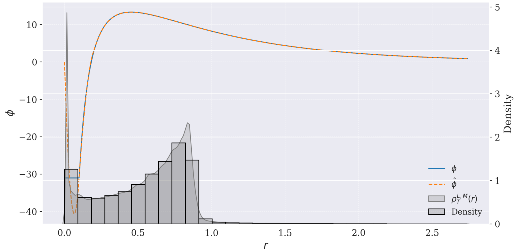


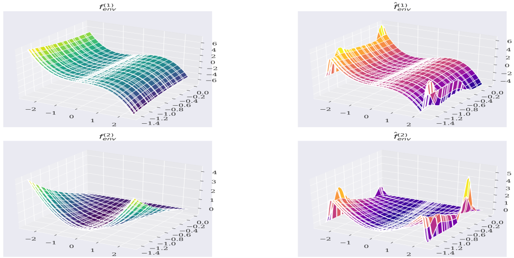


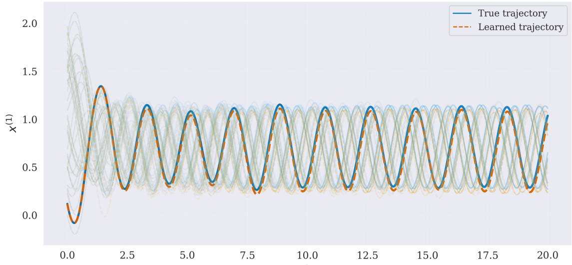




SPP learning: non parametric







Conclusion

- We demonstrated that modeling collective dynamics has applications across various fields.
- This project focused on learning collective dynamics from observations with minimal prior model knowledge.
- We extended an existing algorithm for learning interactions in collective dynamics by incorporating pairwise distances along with an environmental force term.
- We first tested the parametric algorithm using simulated data from the Kuramoto system, the SPP system, and the 1D Cucker-Smale system.
- We then introduced a non-parametric variational algorithm and demonstrated its performance on the same systems.

Future Work

- Compare parametric and non-parametric approaches.
- Determine the minimum number of trajectories M required for convergence for each technique.
- Optimize model hyperparameters.

Thank you!

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